Euro Interest Rates and Euro Exchange Rates

A Time-Series Analysis against the US Dollar, the Japanese Yen and the Pound Sterling

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Abstract

This paper studies the joint dynamics of a euro short-term interest rate and the exchange rate of the euro against three major currencies over 1999–2024: the US dollar, a global reserve currency; the Japanese yen, a safe-haven currency issued under ultra-accommodative monetary policy; and the pound sterling, the currency of a non-euro European economy. All four series are integrated of order one according to sequential ADF and KPSS tests. A Box–Jenkins analysis selects an AR(1) for the first-differenced interest rate, which passes residual autocorrelation and ARCH tests and is used for short-term forecasting. In the multivariate analysis, a VAR(1) in first differences is selected by all information criteria. Granger causality tests show that sterling movements help predict the rest of the system, an effect that runs through the other exchange rates rather than through the interest rate. Impulse responses from the VAR and from local projections show that an interest rate shock is persistent but has no statistically significant effect on the three exchange rates at this frequency. Johansen tests find no cointegration, so the VAR in differences is preferred to a VECM. Because foreign interest rates are not included, the analysis is a first step towards a test of interest rate parity rather than a full test.

1 Introduction

The interaction between interest rates and exchange rates is a central question in international macroeconomics. It links real-economy issues to financial ones: interest rate differentials create arbitrage opportunities and shape capital flows, and through them the exchange rate, which in turn affects trade, imported inflation and financial conditions. For a central bank, understanding how its interest rate decisions transmit to the exchange rate is a key part of understanding monetary policy transmission.

This paper examines the relationship between a euro short-term interest rate and the exchange rate of the euro against three major currencies, chosen for their contrasting roles:

- the **US dollar (USD)**, the world's main reserve currency, whose movements are driven by Federal Reserve decisions and global financial flows;
- the **Japanese yen (JPY)**, often seen as a safe-haven currency, issued by a central bank that has followed an ultra-accommodative policy for most of the period, in an economy marked by low growth and low inflation;

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- the **pound sterling (GBP)**, the currency of a European economy outside the euro area, which allows us to assess the influence of euro area monetary conditions on a European currency outside its direct perimeter.

The natural theoretical benchmark is uncovered interest rate parity (UIP). Letting s_t denote the log of the number of foreign currency units per euro, UIP states that

$$\mathbb{E}_t \left[\Delta s_{t+1}\right] = i_t^* - i_t, \tag{1}$$

where i_t is the euro interest rate and i_t^* the foreign interest rate: the euro is expected to depreciate against a currency that offers a higher return. A large literature documents that UIP fails at short horizons, and that monetary policy shocks can have delayed and persistent effects on exchange rates.

This study should be read as a first step towards an analysis of interest rate parity. Only the euro interest rate is available in our dataset; foreign interest rates, which a full test of equation (1) requires, are not included. We instead focus on a comparison across currency zones and rely on recent data, which allows us to interpret the results in light of the post-pandemic tightening cycle.

The paper proceeds in two steps. Section 2 presents the data. Section 3 carries out the univariate analysis: unit root tests, identification and estimation of an ARMA model for the interest rate, and forecasts. Section 4 turns to the multivariate analysis: a VAR model, Granger causality tests, impulse responses from the VAR and from local projections, and cointegration tests. Section 5 concludes and outlines extensions.

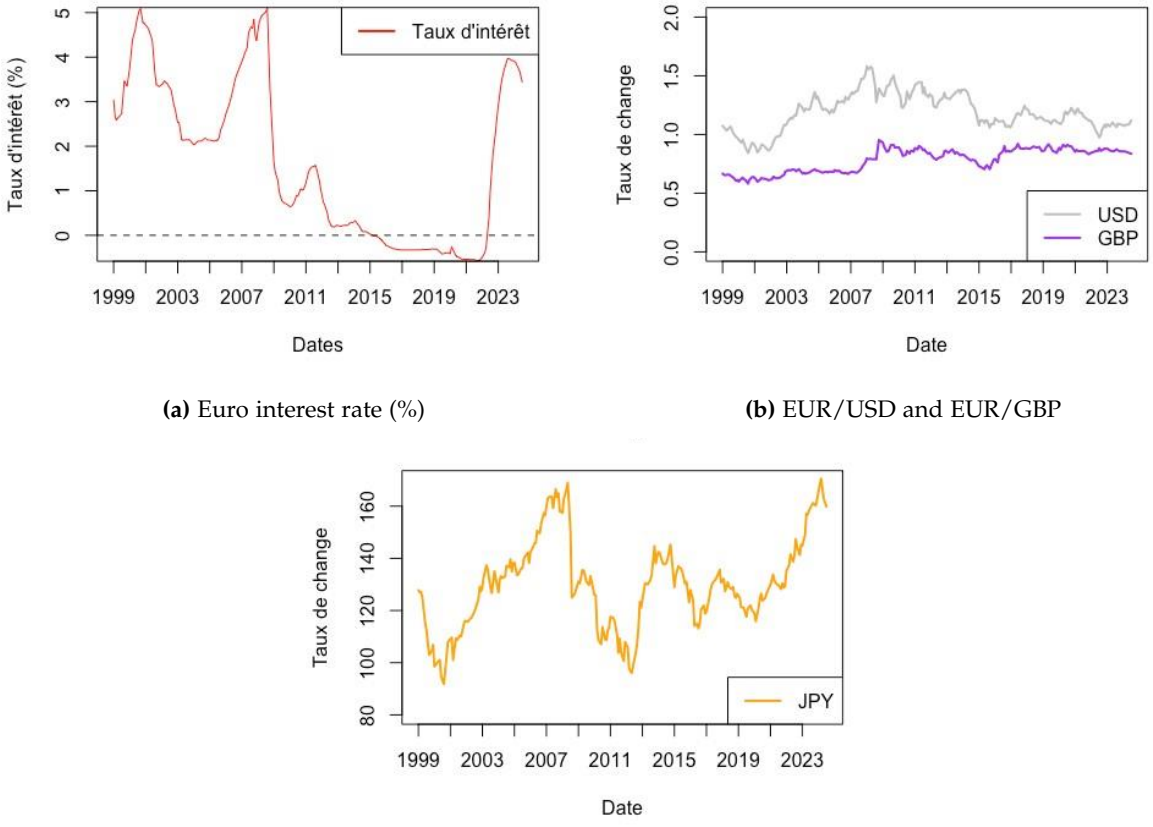
2 Data and descriptive analysis

The dataset contains four series from 1999, the year the euro was introduced, to 2024: a euro short-term interest rate, in percent, and the bilateral exchange rates of the euro against the US dollar, the Japanese yen and the pound sterling, expressed as foreign currency units per euro (an increase is a euro appreciation). After alignment, the sample contains 219 observations. The data are taken from the ECB statistical database.

Figure 1 shows the four series. The interest rate tracks the successive phases of euro area monetary policy. Starting around 3% in 1999, it peaks close to 5% in 2000 and again in 2008, when inflation pressures were high. It then falls sharply to around 1% in 2009 after the global financial crisis, remains low during the sovereign debt crisis, and becomes negative from 2015 onwards, reaching about -0.5% during the negative interest rate period. It rises rapidly from 2022 to around 4% as the ECB tightens to fight inflation, before starting to decline in 2024.

The EUR/USD rate falls around the 2008 and 2020 crises, consistent with the dollar's safe-haven role in periods of global uncertainty. The EUR/GBP rate is less volatile, apart from a sharp sterling depreciation in 2008 and renewed sterling weakness after the 2016 Brexit referendum. The EUR/JPY rate falls sharply after 2008 and during the euro area sovereign debt crisis, reflecting the yen's safe-haven status. It then rises markedly from 2013, when the Bank of Japan launched its quantitative and qualitative easing programme, and again after 2021, as the Bank of Japan maintained its ultra-accommodative stance while other central banks tightened.

Figure 2 reports the correlograms. For all four series, the autocorrelation function decays

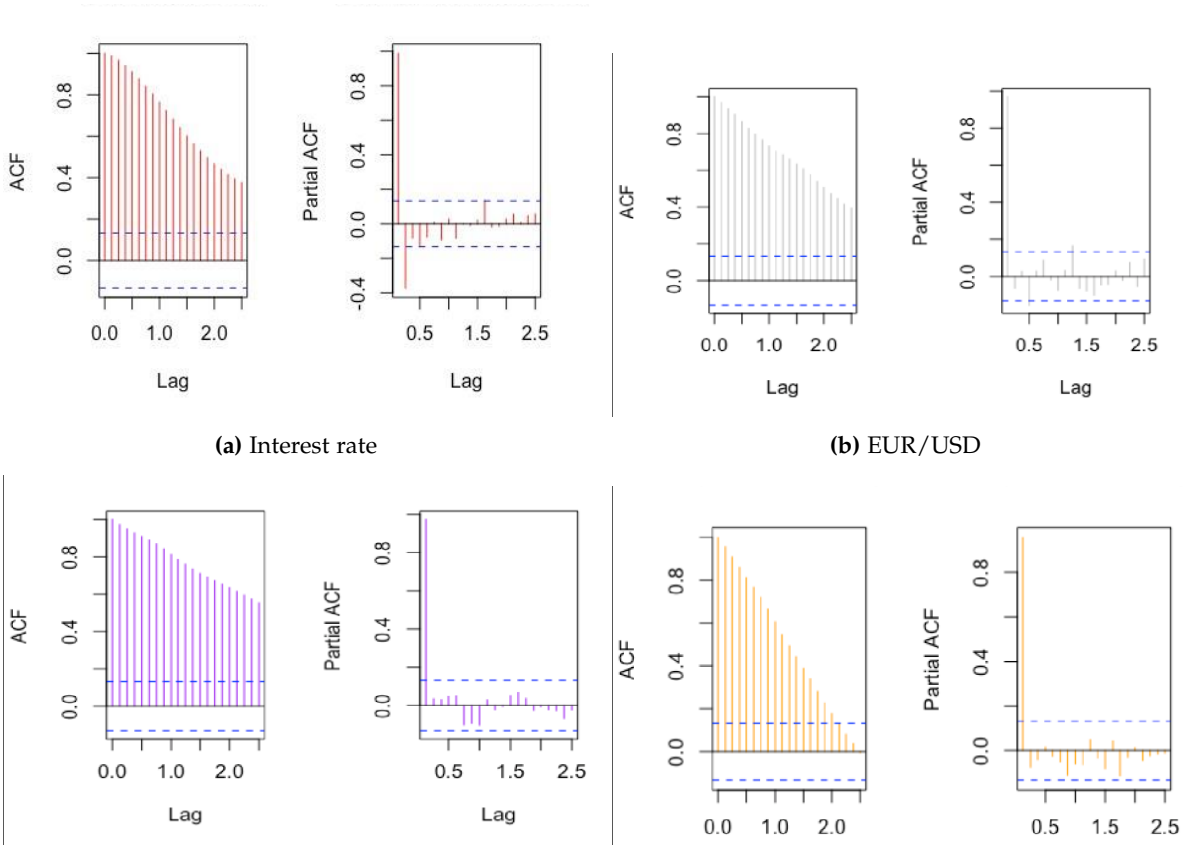


(a) Euro interest rate (%)

(b) EUR/USD and EUR/GBP

(c) EUR/JPY

Figure 1: The four series, 1999–2024. Exchange rates are expressed as foreign currency units per euro.



(a) Interest rate

(b) EUR/USD

(c) EUR/GBP

(d) EUR/JPY

Figure 2: Autocorrelation (left of each panel) and partial autocorrelation (right) functions of the series in levels. Dashed lines are 95% confidence bands. Lags are expressed in units of the time-series frequency used in R.

slowly as the lag increases, which is typical of a non-stationary, highly persistent process: past values strongly influence future values. The partial autocorrelation functions of the exchange rates show a single large spike at the first lag followed by insignificant values, a pattern characteristic of an AR(1) process with a root close to one. For the interest rate, the second partial autocorrelation is also significant, which suggests richer short-run dynamics. These patterns call for formal unit root tests.

3 Univariate analysis

3.1 Unit root tests

We test for a unit root with the augmented Dickey–Fuller (ADF) test, based on the regression

$$\Delta X_t = f_t + \rho X_{t-1} + \sum_{j=1}^{p} \gamma_j \Delta X_{t-j} + \varepsilon_t, \qquad H_0 : \rho = 0 \quad \text{vs.} \quad H_1 : \rho < 0, \tag{2}$$

where f_t is the deterministic component. We follow the standard sequential strategy, starting from the most general model and moving to more restricted ones: model M3 includes a constant and a linear trend ($f_t = c + bt$), model M2 a constant only ($f_t = c$), and model M1 no deterministic term. At each step, we test the unit root using the appropriate Dickey–Fuller critical values, then test the significance of the highest-order deterministic term using the Dickey–Fuller tables, and move to the next model if it is not significant. The number of lagged differences p is selected by AIC.

Table 1: Sequential ADF unit root tests (series in levels)

	M3: constant and trend		M2: constant		M1: none
Series	τ_3	t(trend)	τ_2	t(constant)	τ_1
Interest rate	-2.112	-0.900	-2.009	1.370	-1.467
EUR/USD	-1.877	-0.715	-1.858	1.840	-0.255
EUR/JPY	-2.457	0.048	-2.580	2.494	-0.816
EUR/GBP	-2.829	-2.521	-1.310	1.224	-0.765
5% critical value	-3.43	3.12	-2.88	2.84	-1.95

Notes: τ statistics are compared with the Dickey–Fuller critical values; the t statistics of the deterministic terms are compared in absolute value with the Dickey–Fuller tables. Lag length selected by AIC (three lags for the interest rate, one lag for the exchange rates).

Table 1 summarises the results. For every series, the ADF statistic in model M3 is above the 5% critical value, so the null of a unit root is not rejected, and the trend is never significant. The same holds in model M2, where the constant is never significant. In model M1, the statistics remain above the 5% critical value of -1.95. The sequential strategy therefore concludes that the four series contain a unit root, without drift: they are integrated of order one, I(1).

As a robustness check, we use the KPSS test, whose null hypothesis is the reverse of the ADF test: stationarity. A rejection of the null in levels, and no rejection in first differences, is therefore consistent with the ADF results.

Table 2 confirms these conclusions. In levels, the null of stationarity is rejected at the 5%

Table 2: KPSS stationarity tests

Series	Levels	First differences
Interest rate	2.096	0.209
EUR/USD	0.786	0.095
EUR/JPY	0.468	0.103
EUR/GBP	3.244	0.071
Critical values: 0.347 (10%), 0.463 (5%), 0.574 (2.5%), 0.739 (1%)		

Notes: KPSS test with a constant (level stationarity) and a short lag truncation. H_0: the series is stationary.

level for all series; for the yen, the rejection is borderline (0.468 against a critical value of 0.463). In first differences, all statistics lie well below the 10% critical value. All four series are therefore I(1) and become stationary after first differencing, which we use in the rest of the analysis.

3.2 ARMA model for the interest rate

We model the first-differenced interest rate, Δi_t, following the Box–Jenkins methodology: identification, estimation and validation.

Identification. The correlograms of Δi_t (Figure 3, left) show an autocorrelation function that decays like a damped exponential, characteristic of an autoregressive process, and a partial autocorrelation function with a single large spike at the first lag. This points to an AR(1). We confirm this by estimating all ARMA(p, q) models with $p, q \leq 4$ by maximum likelihood and comparing their information criteria (Table 3). The BIC is minimised by the AR(1). The AIC is minimised by the AR(3), but the AR(1) is its second-best specification, with a very close value. Given the parsimony principle, we retain the AR(1).

Table 3: Information criteria for ARMA(p, q) models of Δi_t

	$q=0$	$q=1$	$q=2$	$q=3$	$q=4$
Panel A: AIC					
$p=0$	-11.08	-70.81	-75.32	-75.09	-81.04
$p=1$	-81.24	-79.33	-80.89	-79.20	-79.33
$p=2$	-79.28	-78.10	-79.04	-76.99	-77.38
$p=3$	$\mathbf{-82.10}$	-80.19	-79.96	-78.63	-77.40
$p=4$	-80.27	-78.79	-78.04	-75.96	-74.80
Panel B: BIC					
$p=0$	-4.31	-60.66	-61.78	-58.17	-60.73
$p=1$	$\mathbf{-71.09}$	-65.79	-63.97	-58.89	-55.64
$p=2$	-65.74	-61.18	-58.74	-53.30	-50.30
$p=3$	-65.18	-59.88	-56.27	-51.56	-46.94
$p=4$	-59.96	-55.10	-50.96	-45.50	-40.96

Notes: Models estimated by maximum likelihood with a constant. The minimum of each criterion is in bold.

Estimation and validation. Table 4 reports the estimates. In the AR(1), the autoregressive coefficient is 0.534 and highly significant, while the constant is not. The estimated process is stationary: the inverse root of the AR polynomial lies well inside the unit circle (Figure 3, right). To check that the model is not under-specified, we add an AR(2) term and an MA(1) term in turn. Neither is significant, and the MA term reduces the precision of the AR coefficient, which confirms the AR(1) specification.

Table 4: ARMA estimates for Δi_t

	AR(1)	AR(2)	ARMA(1,1)
AR(1) coefficient	0.534^{***}	0.541^{***}	0.494^{***}
	(0.058)	(0.068)	(0.148)
AR(2) coefficient		-0.013	
		(0.068)	
MA(1) coefficient			0.055
			(0.184)
Constant	-0.001	-0.001	-0.001
	(0.029)	(0.028)	(0.028)
$\hat{\sigma}^2$	0.0395		
Log-likelihood	43.62		
AIC / BIC	-81.24 / -71.09		
Residual diagnostics for the AR(1)			
Ljung–Box Q^* (16 lags)	18.96 $[p = 0.216]$		
ARCH LM (8 lags)	6.17 $[p = 0.629]$		
Jarque–Bera	13,384 $[p < 0.001]$		

Notes: Maximum likelihood estimates; standard errors in parentheses. *, ** and *** denote significance at the 10%, 5% and 1% levels.

The residual diagnostics support the model. The Ljung–Box test does not reject the absence of residual autocorrelation ($p = 0.22$), and Engle's ARCH test does not detect conditional heteroskedasticity up to order 8 ($p = 0.63$). The Jarque–Bera test strongly rejects normality: the residuals have fat tails, driven by a few very large rate movements such as the sharp cuts of late 2008. This is a common feature of financial series and does not affect the consistency of the estimates, so we retain the AR(1).

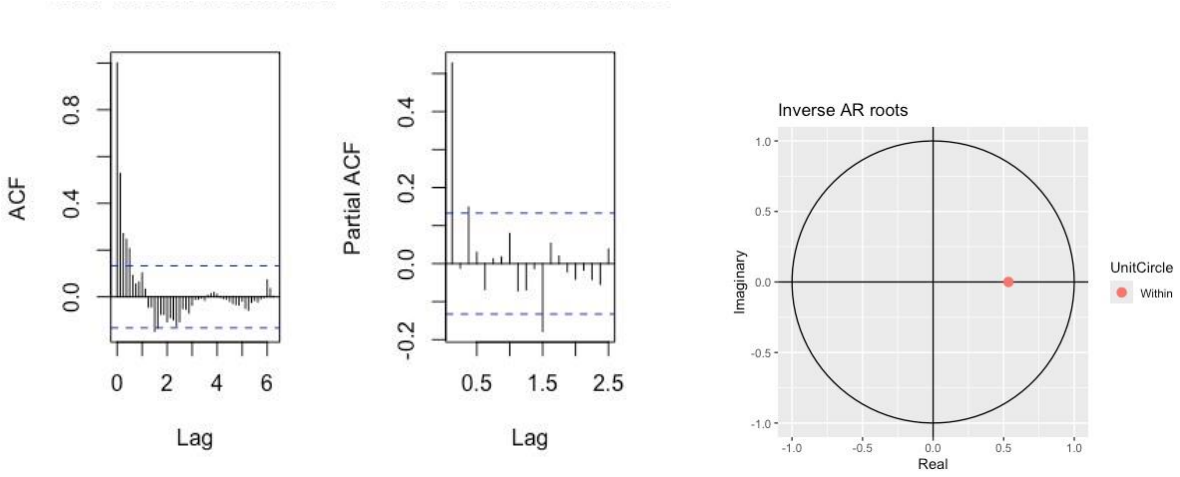
3.3 Forecasts

We use the AR(1) to forecast the interest rate three periods ahead. The forecast $\hat{Y}_T(h) = \mathbb{E}\left(Y_{T+h} \mid \mathcal{I}_T\right)$ uses the information set $\mathcal{I}_T = \{Y_1, \ldots, Y_T\}$. Forecasts of the differenced series describe changes rather than levels, which are needed for economic interpretation. We therefore recover forecasts of the level by cumulating the forecast changes from the last observed value, which amounts to forecasting with an ARIMA(1,1,0) model.

Table 5: AR(1) forecasts of the interest rate

Horizon	$h = 1$	$h = 2$	$h = 3$
Forecast change, $\Delta\hat{i}_{T+h}$ (pp)	-0.135	-0.073	-0.039
Forecast level, $\hat{i}_{T+h}$ (%)	3.299	3.227	3.187

Table 5 shows that the model forecasts a gradual decline in the interest rate, with smaller



(a) ACF and PACF of Δi_t

(b) Inverse root of the estimated AR(1)

Figure 3: Identification and stability of the AR(1) model for the first-differenced interest rate.

decreases at each horizon as the AR(1) dynamics fade out. This is consistent with the easing cycle that started in 2024. Being a purely univariate model, the AR(1) only extrapolates the recent momentum of the series and does not account for the inflation outlook or forward guidance.

4 Multivariate analysis

4.1 VAR model

To study the interactions between the four variables, we estimate a vector autoregression on the stationary first-differenced series, $y_t = (\Delta i_t, \Delta e_t^{USD}, \Delta e_t^{GBP}, \Delta e_t^{JPY})'$, without deterministic terms since the series have no drift. Table 6 shows that all four information criteria select one lag.

Table 6: VAR lag order selection

Lags	1	2	3	4	5	6	7	8
AIC	$\mathbf{-15.09}$	-14.99	-15.00	-14.93	-14.90	-14.82	-14.70	-14.63
HQ	$\mathbf{-14.99}$	-14.79	-14.69	-14.52	-14.38	-14.20	-13.98	-13.80
SC	$\mathbf{-14.83}$	-14.48	-14.24	-13.91	-13.63	-13.29	-12.92	-12.59
FPE ($\times 10^{-7}$)	**2.80**	3.08	3.05	3.29	3.39	3.69	4.14	4.48

Table 7 reports the VAR(1) estimates. The interest rate equation is dominated by its own lag, with a coefficient of 0.536, very close to the univariate AR(1) estimate; lagged changes in the yen exchange rate are significant at the 5% level but economically very small. In the exchange rate equations, the explanatory power is low, as is usual for exchange rates. The most notable result is that lagged changes in EUR/GBP help explain the subsequent changes in EUR/USD and EUR/JPY.

The VAR is stable: all eigenvalues of the companion matrix are smaller than one in modulus. The adjusted portmanteau test does not reject the absence of residual autocorrelation ($p = 0.12$), consistent with the diagnostics of the univariate AR(1). We therefore retain the VAR(1).

Table 7: VAR(1) estimates (first-differenced series)

	Dependent variable			
	Δi_t	Δe_t^{USD}	Δe_t^{GBP}	Δe_t^{JPY}
Δi_{t-1}	0.536^{***}	-0.017	-0.004	-0.844
	(0.056)	(0.011)	(0.006)	(1.320)
Δe_{t-1}^{USD}	-0.649	0.062	-0.024	-1.539
	(0.456)	(0.090)	(0.051)	(10.729)
Δe_{t-1}^{GBP}	0.014	-0.340^{**}	-0.071	-55.315^{***}
	(0.692)	(0.137)	(0.077)	(16.261)
Δe_{t-1}^{JPY}	-0.007^{**}	0.001	0.0005	0.090
	(0.003)	(0.001)	(0.0004)	(0.081)
R^2	0.330	0.048	0.015	0.068
F statistic (p-value)	26.21 (< 0.001)	2.67 (0.033)	0.81 (0.518)	3.87 (0.005)
Observations	217	217	217	217

Notes: OLS estimates equation by equation, no deterministic terms; standard errors in parentheses. *, ** and *** denote significance at the 10%, 5% and 1% levels. Moduli of the companion matrix eigenvalues: 0.562, 0.118, 0.118, 0.085. Adjusted portmanteau test (10 lags): $\chi^2(144) = 164.41$, $p = 0.117$.

4.2 Granger causality

Including all four variables in the VAR accounts for cross-influences and limits the risk of spurious causality results due to omitted variables. A variable X does not Granger-cause the other variables Y if

$$\mathbb{E}\left(Y_{t+h} \mid Y_t, Y_{t-1}, \ldots, X_t, X_{t-1}, \ldots\right) = \mathbb{E}\left(Y_{t+h} \mid Y_t, Y_{t-1}, \ldots\right). \tag{3}$$

Table 8: Granger causality tests in the VAR(1)

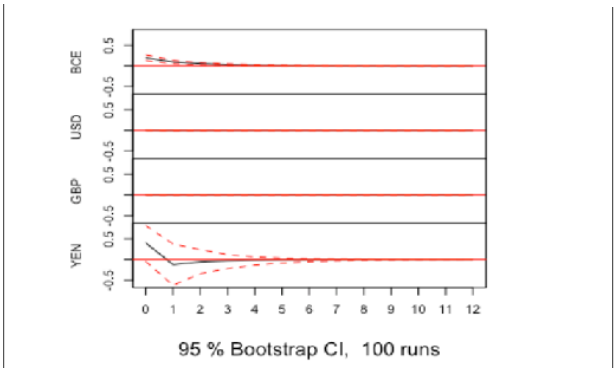
Cause	Effect	F statistic	d.f.	p-value
Δe^{GBP}	Δi, Δe^{USD}, Δe^{JPY}	4.068	(3, 852)	0.007
Δe^{JPY}	Δi, Δe^{USD}, Δe^{GBP}	2.100	(3, 852)	0.099

Notes: H_0: the variable in the first column does not Granger-cause the other variables of the VAR jointly.

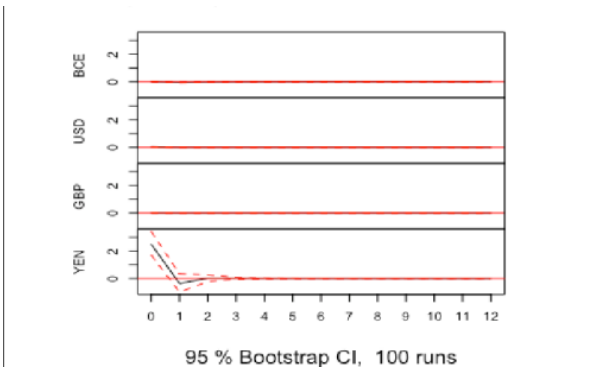
Table 8 shows that the null hypothesis is rejected at the 1% level for sterling: changes in EUR/GBP help predict the rest of the system. The equation-by-equation estimates in Table 7 show where this result comes from. Lagged sterling is significant in the EUR/USD and EUR/JPY equations, but not in the interest rate equation ($p = 0.98$). The predictive content of sterling therefore runs through the other euro exchange rates rather than through euro interest rates. One possible interpretation is that EUR/GBP movements partly reflect euro-specific news, which is then priced into the other euro exchange rates. The yen Granger-causes the system only at the 10% level, an effect that comes mainly from its small but significant coefficient in the interest rate equation.

4.3 Impulse response analysis

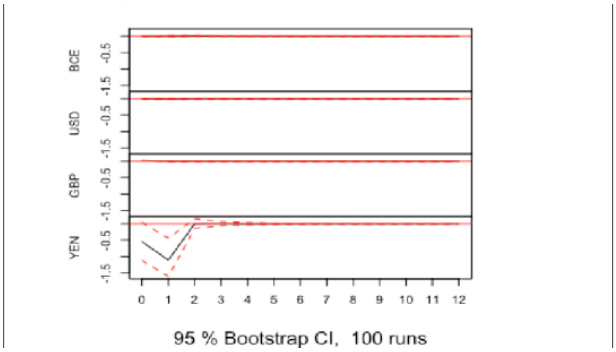
VAR impulse responses. We identify structural shocks with a Cholesky decomposition, ordering the interest rate first: it is assumed to react to exchange rates only with a lag, while exchange rates, as fast-moving asset prices, can react to it within the period. Figure 4 reports the orthogonalised impulse responses with 95% bootstrap confidence intervals. Because the EUR/JPY rate is quoted in yen per euro, its changes are much larger in magnitude than those of the other variables, which compresses the scale of the plots and makes the responses of the other variables hard to read.



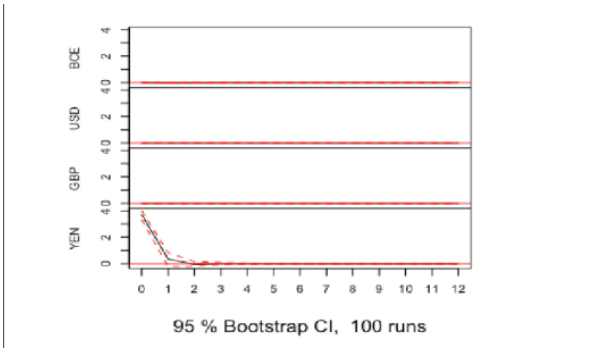
(a) Interest rate shock



(b) EUR/USD shock



(c) EUR/GBP shock



(d) EUR/JPY shock

Figure 4: Orthogonalised impulse responses from the VAR(1), Cholesky ordering (interest rate, USD, GBP, JPY). From top to bottom in each panel: responses of the interest rate (BCE), EUR/USD, EUR/GBP and EUR/JPY over 12 periods. Dashed lines: 95% bootstrap confidence intervals (100 replications).

Local projections. As a complement, we estimate impulse responses with local projections, which regress each variable at horizon h directly on the identified shock and controls:

$$y_{k,t+h} = \alpha_k^h + \beta_k^h \, \varepsilon_{j,t} + \sum_{l=1}^{L} \Gamma_{k,l}^h \, y_{t-l} + u_{k,t+h}, \qquad h = 0, 1, \ldots, 12, \tag{4}$$

where β_k^h is the response of variable k at horizon h to a shock $\varepsilon_{j,t}$ in variable j, identified with the same Cholesky ordering. Local projections are more robust to misspecification of the VAR dynamics at longer horizons, at the cost of less precise estimates. Each response is estimated separately, so the scale of each panel adapts to the variable (Figure 5).

The first row of Figure 5 shows the responses to an interest rate shock. The response of the interest rate itself is positive and decays gradually over about ten periods, reflecting

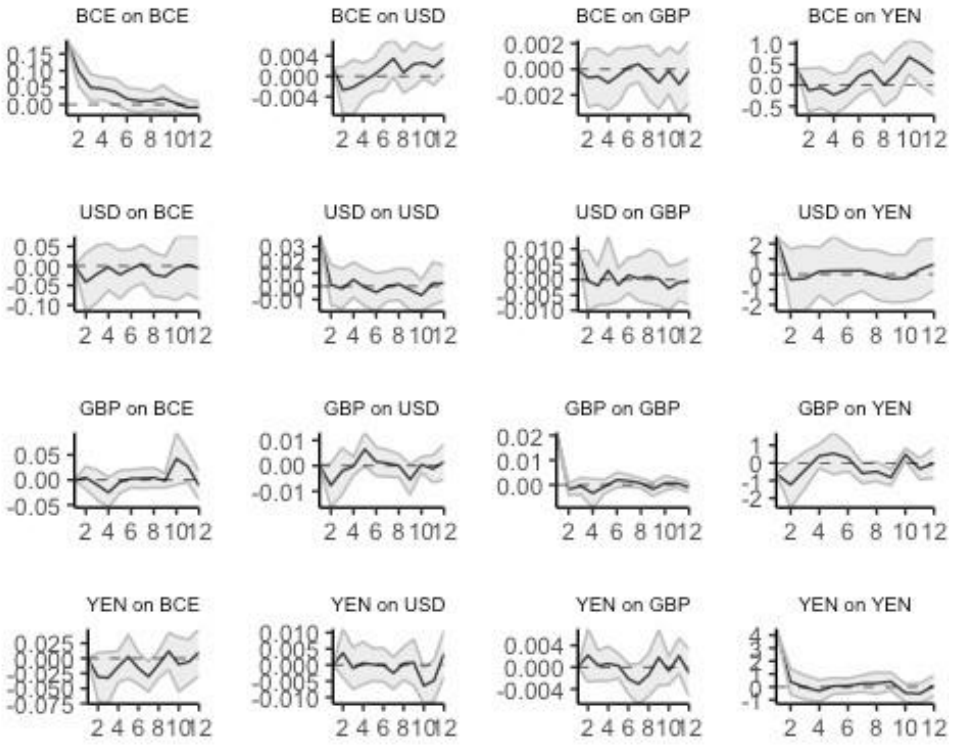


Figure 5: Impulse responses from local projections over 12 periods. Each panel reads "shock on response"; "BCE" denotes the interest rate. Shaded areas: 95% confidence intervals.

the persistence of interest rate movements: the effect of a shock fades out progressively as the economy adjusts. By contrast, the responses of the three exchange rates are small and their confidence intervals include zero at almost all horizons. At this frequency and without identifying genuine monetary policy surprises, we find no statistically significant effect of interest rate shocks on the euro exchange rate. The weak reaction of the dollar is consistent with the dominant role of Federal Reserve policy in driving EUR/USD, and that of the yen with the Bank of Japan's focus on domestic objectives, the yen reacting mainly to global rather than European shocks. The other rows show that each exchange rate responds mainly to its own shock, with little persistence, consistent with exchange rates behaving close to random walks.

4.4 Cointegration

Since the four series are I(1), a long-run equilibrium relationship between them could exist, in which case a vector error correction model (VECM) would be more appropriate than a VAR in differences. We test for cointegration with the Johansen trace test:

$$\lambda_{\text{trace}}(r) = -T \sum_{i=r+1}^{n} \ln\left(1 - \hat{\lambda}_i\right), \tag{5}$$

where $\hat{\lambda}_i$ are the estimated eigenvalues and r the cointegration rank under the null. The test requires at least two lags in the VAR in levels, so we use $K = 2$ for the test only, without changing the VAR(1) retained above. We first test each pair formed by the interest rate and

one exchange rate, then all four series jointly.

Table 9: Johansen trace tests

System	H_0	Trace statistic	10%	5%	1%
Rate – USD	$r = 0$	15.46	15.66	17.95	23.52
	$r \leq 1$	6.20	6.50	8.18	11.65
Rate – GBP	$r = 0$	6.38	15.66	17.95	23.52
	$r \leq 1$	2.17	6.50	8.18	11.65
Rate – JPY	$r = 0$	10.84	15.66	17.95	23.52
	$r \leq 1$	2.98	6.50	8.18	11.65
All four series	$r = 0$	33.38	45.23	48.28	55.43
	$r \leq 1$	18.54	28.71	31.52	37.22
	$r \leq 2$	7.78	15.66	17.95	23.52
	$r \leq 3$	2.63	6.50	8.18	11.65

Notes: Trace test with $K = 2$ lags in the VAR in levels. Eigenvalues of the four-variable system: 0.066, 0.048, 0.023, 0.012.

Table 9 shows that the null of no cointegration is never rejected at the 5% level, either for the pairs or for the four-variable system, in which multi-variable relationships invisible in pairwise tests could have appeared. The closest case is the interest rate and the dollar, where the statistic for $r = 0$ is just below the 10% critical value. In the absence of cointegration, a VECM is not justified, and the VAR in first differences remains the most appropriate representation.

5 Conclusion

This paper analysed the dynamics of a euro short-term interest rate and of the euro exchange rate against the dollar, the yen and sterling. All series are strongly persistent and integrated of order one, as shown by ADF and KPSS tests, and become stationary after first differencing. An AR(1) adequately describes the dynamics of the differenced interest rate and forecasts a gradual decline in rates, consistent with the easing cycle under way at the end of the sample. In the multivariate analysis, a stable VAR(1) shows that sterling movements help predict the other euro exchange rates, while interest rate shocks have persistent effects on the rate itself but no significant effect on the three exchange rates at this frequency. There is no evidence of cointegration.

These results should be placed in a global context in which central banks face persistent inflation, geopolitical uncertainty and imbalances in trade and financial flows. In such an environment, the ECB cannot ignore the effects of its decisions beyond the euro area, on partner economies and on the currencies most closely linked to the euro.

The analysis has several limitations, which suggest natural extensions. First, foreign interest rates are not included, so equation (1) is not tested directly; adding Federal Reserve, Bank of Japan and Bank of England rates would allow a proper test of interest rate parity based on interest rate differentials. Second, changes in a short-term rate mix monetary policy shocks with the systematic response of policy to the economy. Identifying genuine monetary policy surprises from high-frequency movements in market rates around ECB announcements, and using them in local projections, would give the impulse responses a causal interpretation. Third, part of the sample is close to the effective lower bound, where policy operated mainly

through asset purchases and forward guidance; a shadow rate would better capture the monetary stance in that period.

A R code

The analysis was carried out in R with the packages `urca`, `forecast`, `lmtest`, `FinTS`, `tseries`, `vars` and `lpirfs`. The full script is reproduced below and is available in the `code/` folder of the repository.

```r
# ================================================================================
# Euro Interest Rates and Euro Exchange Rates: a time-series analysis
# Oscar Combe - Universite Paris Dauphine-PSL (December 2024)
# ================================================================================

library(urca) # ADF, KPSS and Johansen tests
library(forecast) # ARIMA estimation and forecasts
library(lmtest) # coefficient tests
library(FinTS) # ARCH LM test
library(tseries) # Jarque-Bera test
library(vars) # VAR estimation, causality, IRFs
library(lpirfs) # local projections
library(ggplot2)

# --- 0. Data ------------------------------------------------------------------
# Expected files (see data/README.md):
# data/interest_rate.csv : columns DATE, RATE
# data/exchange_rates.csv : columns Date, USD, JPY, GBP (foreign currency per euro)
base_rate <- read.csv("data/interest_rate.csv")
base_fx <- read.csv("data/exchange_rates.csv")

rate <- base_rate$RATE
usd <- base_fx$USD
jpy <- base_fx$JPY
gbp <- base_fx$GBP

# --- 1. Plots -----------------------------------------------------------------
dates_rate <- as.Date(base_rate$DATE)
plot(dates_rate, rate, type = "l", col = "red", main = "Euro interest rate",
     xlab = "Date", ylab = "Interest rate (%)", xaxt = "n", yaxs = "i")
abline(h = 0, lty = 2)
axis.Date(1, at = seq(min(dates_rate), max(dates_rate), by = "2 years"), format = "%Y")

dates_fx <- as.Date(base_fx$Date)
plot(dates_fx, usd, type = "l", col = "grey", lwd = 2, ylim = c(0, 2), xaxt = "n",
     main = "EUR/USD and EUR/GBP", xlab = "Date", ylab = "Exchange rate")
lines(dates_fx, gbp, col = "purple", lwd = 2)
legend("bottomright", legend = c("USD", "GBP"), col = c("grey", "purple"), lwd = 2)
axis.Date(1, at = seq(min(dates_fx), max(dates_fx), by = "2 years"), format = "%Y")

plot(dates_fx, jpy, type = "l", col = "orange", lwd = 2, ylim = c(80, 170), xaxt = "n",
     main = "EUR/JPY", xlab = "Date", ylab = "Exchange rate")
axis.Date(1, at = seq(min(dates_fx), max(dates_fx), by = "2 years"), format = "%Y")

# Correlograms (same code for each series)
par(mfrow = c(1, 2))
acf(rate, lag.max = 20, main = "ACF (rate)", col = "red")
pacf(rate, lag.max = 20, main = "PACF (rate)", col = "red")
par(mfrow = c(1, 1))

# --- 2. Unit root tests -------------------------------------------------------
# Sequential ADF strategy: M3 (trend) -> M2 (drift) -> M1 (none)
summary(ur.df(rate, type = "trend", lags = 4, selectlags = "AIC"))
```

```r
summary(ur.df(rate, type = "drift", lags = 4, selectlags = "AIC"))
summary(ur.df(rate, type = "none", lags = 4, selectlags = "AIC"))

# KPSS (H0: stationarity), in levels and first differences
summary(ur.kpss(rate, type = "mu", lags = "short"))
summary(ur.kpss(diff(rate), type = "mu", lags = "short"))

fx <- data.frame(USD = usd, JPY = jpy, GBP = gbp)
for (s in names(fx)) {
  cat("\n=== Series:", s, "===\n")
  for (m in c("trend", "drift", "none")) {
    print(summary(ur.df(fx[[s]], type = m, selectlags = "AIC")))
  }
  print(summary(ur.kpss(fx[[s]], type = "mu", lags = "short")))
  print(summary(ur.kpss(diff(fx[[s]]), type = "mu", lags = "short")))
}

# --- 3. ARMA model for the differenced interest rate (Box-Jenkins) -------------
d_rate <- diff(rate)

par(mfrow = c(1, 2))
acf(d_rate, lag.max = 50, main = "ACF of the differenced rate")
pacf(d_rate, lag.max = 20, main = "PACF of the differenced rate")
par(mfrow = c(1, 1))

# Information criteria for ARMA(p,q), p,q <= 4
mat_aic <- mat_bic <- matrix(NA, 5, 5,
                             dimnames = list(paste0("p=", 0:4), paste0("q=", 0:4)))
for (p in 0:4) for (q in 0:4) {
  fit <- Arima(d_rate, order = c(p, 0, q), method = "ML")
  mat_aic[p + 1, q + 1] <- fit$aic
  mat_bic[p + 1, q + 1] <- fit$bic
}
print(mat_aic); print(mat_bic)

# Estimation and validation
AR1 <- Arima(d_rate, order = c(1, 0, 0))
summary(AR1); coeftest(AR1)
autoplot(AR1) # inverse AR roots

coeftest(Arima(d_rate, order = c(2, 0, 0))) # over-fitting: AR(2)
coeftest(Arima(d_rate, order = c(1, 0, 1))) # over-fitting: ARMA(1,1)

checkresiduals(AR1) # Ljung-Box
ArchTest(AR1$residuals, lags = 8) # Engle ARCH LM test
jarque.bera.test(AR1$residuals) # normality

# --- 4. Forecasts --------------------------------------------------------------
fc_diff <- forecast(AR1, h = 3)
fc_level <- cumsum(c(tail(rate, 1), fc_diff$mean))[-1] # back to levels
print(fc_diff$mean); print(fc_level)

# --- 5. VAR on first differences -----------------------------------------------
endogen <- ts.intersect(BCE = diff(rate), USD = diff(usd),
                        GBP = diff(gbp), YEN = diff(jpy))

VARselect(endogen, lag.max = 8, type = "none")$selection
VAR1 <- VAR(endogen, p = 1, type = "none")
summary(VAR1)
roots(VAR1) # stability
serial.test(VAR1, lags.pt = 10, type = "PT.adjusted") # portmanteau test

# Granger causality
for (v in colnames(endogen)) print(causality(VAR1, cause = v)$Granger)
```

```r
# --- 6. Impulse responses ----------------------------------------------------
# Cholesky ordering: rate, USD, GBP, JPY
for (v in colnames(endogen)) {
  plot(irf(VAR1, impulse = v, n.ahead = 12, ortho = TRUE, ci = 0.95))
}

irf_lp <- lp_lin(endog_data = as.data.frame(endogen), lags_endog_lin = 1,
                 trend = 0, shock_type = 0, confint = 1.96, hor = 12,
                 adjust_se = TRUE)
plot(irf_lp)

# --- 7. Johansen cointegration tests (levels) --------------------------------
pairs <- list(rate_USD = cbind(rate, usd),
              rate_GBP = cbind(rate, gbp),
              rate_JPY = cbind(rate, jpy))
for (n in names(pairs)) {
  cat("\n=== Pair:", n, "===\n")
  print(summary(ca.jo(pairs[[n]], type = "trace", K = 2,
                      ecdet = "none", spec = "transitory")))
}
summary(ca.jo(cbind(rate, usd, gbp, jpy), type = "trace", K = 2,
              ecdet = "none", spec = "transitory"))
```